PROVEN

From lab to refinery.

7/7
Factory Acceptance Test items passed · second-generation GLD, laboratory

8
Cluster Heads in a multi-hop mesh, failover tested · ITB campus

2025
Field trial at RU VII Kasim, Nov–Dec · first-generation GLD


Equipment check with HSE personnel


Pole installation, witnessed by refinery staff


Installation on a solar-powered pole


GLD installed in the tank area


Data monitoring from the control room


Team coordination on site

Field trial at RU VII Kasim · November–December 2025 · first-generation GLD.

SPECIFICATIONS

Sensors	Multiple metal-oxide gas sensors
Intelligence	Dual-branch CNN, 8-bit, on-device
Communication	LoRa, private multi-hop mesh
Power supply	24 VDC
Validated gases	LPG, H_2
In development	CO_2 (data available) · CH_4, CO, NH_3, benzene (targets) · models trained on site

PERTAMINA PATRA NIAGA

Pilot: RU IV Cilacap

GAS LEAK DETECTION SYSTEM

GLD V2

Intelligent, wireless multi-gas leak detection for refineries.



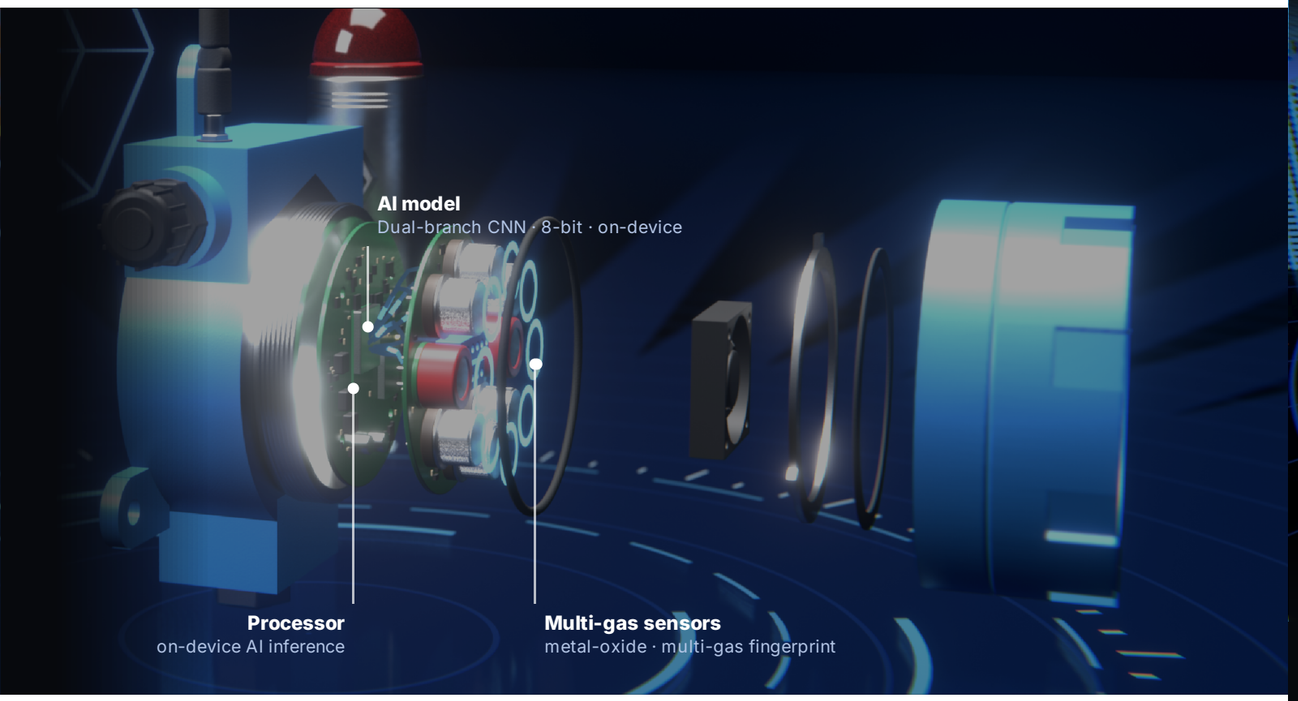
PT Pertamina Patra Niaga (Fungsi Reliability)
LAPI Ganesha Utama · Institut Teknologi Bandung

3D illustration

THE DEVICE

Multi-gas sensors. One device.

Each GLD reads the response pattern of multiple gas sensors, and on-device AI identifies the gas. No server round-trip, no waiting.



3D illustration of GLD V2, exploded view.

Gas fingerprint

Metal-oxide sensors form a distinct pattern for each gas.

Edge AI

A dual-branch 8-bit CNN runs directly on the microcontroller.

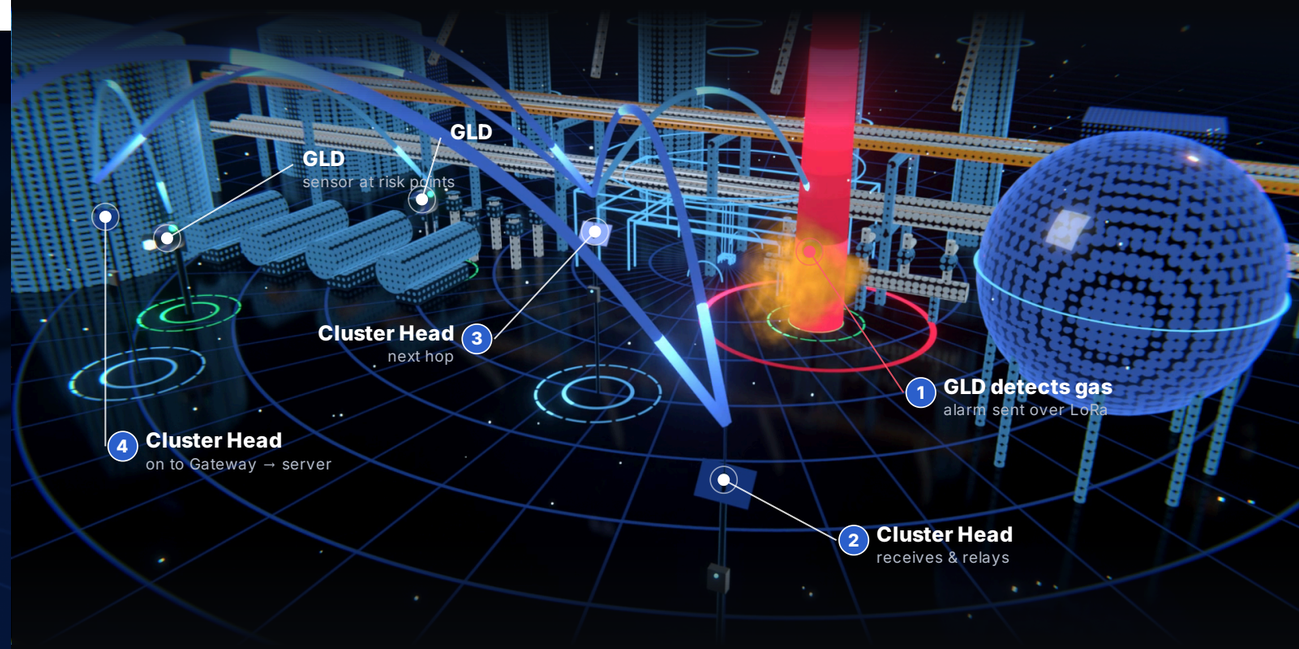
Local alarm

The on-device beacon lights up the moment gas is detected.

THE SYSTEM

Leak detected. Operators know instantly.

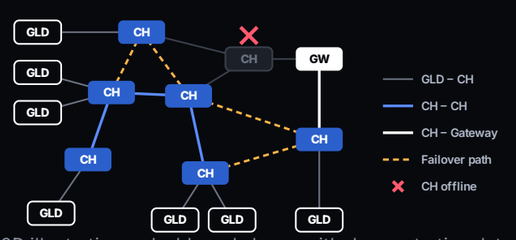
Data travels over a private LoRa mesh. No data cabling, no internet. If one path drops, the network reroutes itself.



GLD → Cluster Head → Gateway → Local server → Dashboard

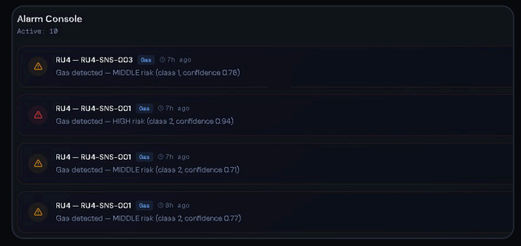
Self-healing network

If a Cluster Head goes offline, data is automatically rerouted through another path.



Instant push alarms

Alarms bypass the polling cycle; operators see location and risk level at once.



3D illustrations · dashboard shown with demonstration data.